Unified Failure Strategy

Epic CDS-121526. Service and environment failure strategy for Unified (V1) pipelines, and failure handling for multi-deployment, run against a local Harness stack and backed by unit tests.

SERVICE & ENVIRONMENT

`on-failure` on a service or environment: retry, abort, fail, pipeline-rollback and (environment only) stage-rollback, in object and bare-string form.

MULTI-DEPLOYMENT

Per-child failure isolation, `skip-queued` / `run-all`, and `max-concurrency`, also set through runtime inputs.

25	25	18	5	2
TEST CASES	PASS	PIPELINE	UNIT TESTS	BOTH

Pipeline cases were executed as real V1 pipelines through the pipeline API (create, execute, read the execution graph), not only written. Retry attempts were read back from `interruptHistories` on the failing node. Validation and K8s fail-fast cases are covered by unit tests because they are decided before or outside a pipeline run.

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01 What each test checks

#	Area	What it checks	Level	Result
1	Service retry	Invalid service with `retry` (3 attempts): 3 retries, then the `failure-action` runs.	Pipeline	PASS
2	Service bare strings	`abort`, `fail`, `pipeline-rollback` written as plain strings are accepted and applied.	Pipeline	PASS
3	Retry, bare failure-action	`retry` with `failure-action: abort` as a plain string, at service and environment.	Pipeline	PASS
4	Environment retry	Invalid environment with `retry` (3 attempts): retries are recorded on the Infrastructure node.	Pipeline	PASS
5	Environment bare strings	`abort`, `fail`, `pipeline-rollback`, `stage-rollback` as plain strings are accepted and applied.	Pipeline	PASS
6	Environment stage-rollback	Stage rollback runs the stage's `rollback:` steps after an environment failure.	Pipeline	PASS
7	Error match: wildcard	`errors: [unknown]` catches a "not found" service error at service scope.	Pipeline	PASS
8	Error match: narrow list	`errors: [approval-rejection]` does not catch a "not found" error.	Pipeline	PASS
9	Disallowed actions	Service rejects ignore, stage-rollback, proceed-with-default; environment rejects ignore. NG wording.	Unit test	PASS
10	Retry follow-up action	Retry followed by `ignore` is rejected.	Unit test	PASS
11	No retry in retry	A retry's `failure-action` cannot itself be `retry`, at service and environment.	Unit test	PASS
12	Empty action	An empty action fails with "must specify an action".	Unit test	PASS
13	Multi-service isolation	Two services, one invalid: the invalid child fails alone, the valid child is not affected.	Pipeline	PASS
14	Multi-environment isolation	Two environments, one invalid: same isolation on the environment side.	Pipeline	PASS
15	Skip-queued	`on-failure: skip-queued` with `max-concurrency: 1`: children queued behind the failure are skipped.	Pipeline	PASS
16	Runtime input + K8s fail-fast	Run-all and max-concurrency supplied as runtime inputs; K8s Initialize ignores only "service not found".	Pipeline + Unit	PASS
17	Service rule beats stage rule	Service `abort` wins over a stage `stage-rollback`; the stage rollback steps do not run.	Pipeline	PASS
18	Fallback to stage rule	Service rule for `[approval-rejection]` does not match "service not found", so the stage `stage-rollback` runs.	Pipeline	PASS
19	Pipeline rollback	Service failure in stage 2 with `pipeline-rollback` rolls back the already deployed stage 1.	Pipeline	PASS
20	Inherited not-allowed action	Stage `ignore` with no service or environment rule: the inherited action is applied, not skipped.	Pipeline	PASS
21	Happy path unaffected	Valid service and environment with rules set: Rendering, Templating, Resource lock and steps all succeed.	Pipeline	PASS
22	Valid env, missing infra	Infra id that does not exist in a valid environment: the environment rule applies on the Infrastructure node, as in NG.	Pipeline + Unit	PASS
23	Failure in rollback's infra	Infrastructure deleted while the pipeline runs: the rollback's Infrastructure node fails under the environment rule.	Pipeline	PASS
24	Invalid env with all-infra	`all-infra: true` on an invalid environment: Service succeeds, the Infrastructure node fails under the environment rule.	Pipeline	PASS
25	Stage without service	K8s Initialize builds no implicit steps when the stage has no service.	Unit test	PASS

Pipeline = executed as a V1 pipeline and read back from the execution graph. Unit test = JUnit, run with `make test`. Cases 16 and 22 have a pipeline run and a unit test.

02 Service & environment failure strategy

1 · Service retry, 3 attempts

fixture V01

Setup Service id invalid; `errors: [unknown]`, `retry` attempts 3, interval 10s, `failure-action: abort`.

Expected Three retries, then abort.

Observed: the Service node's `interruptHistories` holds 3 entries of type RETRY, followed by ABORT. Re-run after the review change that restored `consume` for the service output (run `9r3lDHJeQNy3_AB26WffvQ`): same result.

4 · Environment retry, 3 attempts

fixture V02

Setup Environment id invalid; same retry block under `environment.on-failure`.

Expected Same retry behaviour, recorded on the Infrastructure node, since that is where environment errors surface.

Observed: 3 RETRY entries then ABORT on the Infrastructure node; the Service node succeeded.

2, 3, 5 · Bare-string actions

fixtures B-svc-*, B-env-*

Setup The form the UI emits: `action: abort`, `fail`, `pipeline-rollback`, `stage-rollback` (environment only), and `failure-action: abort` under retry.

```
on-failure:
  - errors: [unknown]
    action: pipeline-rollback
```

Observed: every pipeline was created and executed; the action fired at the expected node. `retry` itself stays object-form only.

6 · Environment stage-rollback

fixture R-env-stage-rollback

Setup Invalid environment, `action: stage-rollback`, and a `rollback:` step `Undo` that runs `echo ROLLBACK_RAN`.

Expected The rollback step executes after the environment failure.

Observed: `Execution (Rollback)` ran and the `Undo` step completed.

7, 8 · Which errors a strategy catches

Earlier session

Setup Service not found, once with `errors: [unknown]` and once with `errors: [approval-rejection]`.

Observed: with `unknown` the action fired at service scope (stage Failed, downstream Skipped). With the narrow list the raw not-found error surfaced and no strategy message appeared, which is the same as NG.

03 Validation (unit tests)

These checks run when the run starts, so they are asserted in code. The message matches NG: `Invalid failure strategy action provided for service` (or `environment`).

Test	Input	Expected
9	Service with `ignore`, `stage-rollback`, `proceed-with-default`; environment with `ignore`	Rejected with the NG wording
10	Retry whose `failure-action` is `ignore`	Rejected
11	Retry whose `failure-action` is `retry` (service and environment)	Rejected
12	Action list with no action	"must specify an action"

Class: `UnifiedFailureStrategyHelperV1Test`.

04 Multi-deployment

13, 14 · Failure isolation — `fixtures V05, V06`

Setup Two items with `parallel: true` (service for V05, environment for V06), one id invalid, `action: fail`.

Observed: only the invalid child failed. The valid child ran independently.

15 · skip-queued — `fixture V08`

Setup Three services in order (valid, invalid, valid), `on-failure: skip-queued`, `max-concurrency: 1`.

Observed: the child after the failure did not run.

16 · Runtime inputs (run-all) — `fixture V09`

Setup Same three services; `on-failure` and `max-concurrency` are `<+input>`. Run with `run-all` and concurrency 2 through the overlay inputs.

```
inputs:
  overlay:
    stages:
      - id: ID_V09_runtime_input
        multi-deployment:
          on-failure: run-all
          max-concurrency: 2
```

Observed: the child after the failure still ran, so the input was applied. An earlier attempt with a wrongly shaped body was ignored; the overlay shape above is the one the UI sends.

05 K8s Initialize fail-fast

Test 16 (unit part): `ModulesImplicitStepsConfigHandlerTest`. Three "service not found" message formats are ignored, as before, so the service step can report them. "Connection refused" and "Environment ... not found" are rethrown instead of being swallowed.

Tests 22 and 25 (unit part), same class: "No infrastructure entity found" is recognised and left for the Infrastructure step, other infrastructure errors are not; a stage with no service builds no implicit steps.

06 Priority, fallback and rollback

17 · Service rule beats stage rule

Demo 3 · U1dbgkTITbqgjLyemmH8Qg

Setup Invalid service, service `[all]` → `abort`; stage `stage-rollback` with an `Undo` rollback step.

Observed: Service node Aborted, run Aborted, `Undo` did not run.

18 · Fallback to the stage rule

Demo 4 · 7vXfAuGkQLWYjNwfCbWiMg

Setup Invalid service, service `[approval-rejection]` → `abort`; stage `stage-rollback` with `Undo`.

Observed: the error is "service not found", not an approval rejection, so the service rule does not match. The stage rule ran: `Execution (Rollback)` and `Undo` succeeded, stage Failed.

19 · Pipeline rollback

Demo 6 · Lv4adCscTxulvM821j993w

Setup Stage 1 deploys fine and has an `Undo` rollback step; stage 2 has an invalid service with `pipeline-rollback`.

Observed: stage 2 Failed at the Service node; the rollback run `Ex3ADZ9FTA-P5pDwCJD91A` rolled back stage 1 and `Undo` succeeded.

20 · Inherited not-allowed action

Check 1 · C5N5kSeETTetn-dCdg2FuA

Setup Invalid service, no service or environment rule, stage `ignore`.

Observed: Service and Infrastructure were both marked IgnoreFailed. The inherited `ignore` is applied, not skipped; the spec was corrected to say so.

21 · Happy path

Check 2 · 33xi1puLRh-gRd7m7puVyA

Setup Valid service and environment, both with `abort`.

Observed: Rendering, Templating, Resource Constraint and `Deploy` all succeeded.

07 Missing infrastructure, NG parity

22 · Valid environment, missing infrastructure

Check 4

Setup Valid environment, `deploy-to: INVALID_INFRA_ID`, environment `abort`.

Before fix Run `pvVESBYERrKvHnHh-B8TvA`: failed at K8s Initialize, the environment rule never ran.

Observed after fix: run `ObR65PCCRIWk9lkHLiVX0A` Aborted on the Infrastructure node, Service succeeded. Same as NG.

23 · Failure in the rollback's infrastructure step

Check 3

Setup Stage 1 on a temporary infrastructure with environment `abort`; stage 2 waits 90s then fails with `pipeline-rollback`. The infrastructure is deleted while stage 2 waits.

Before fix Rollback run `vyb0psyNSCml1aTbBs87ug` failed at Initialize.

Observed after fix: run `WwHAlNjQS6CUdQYYb4a2BQ`, rollback run `8ygYLrZVTD2FaifQXJbzTA`: the rollback's Infrastructure node Aborted with "No infrastructure entity found", under the environment rule.

24 · Invalid environment with all-infra

Check 5

Setup `all-infra: true` on an invalid environment; service `fail`, environment `abort`.

Observed: run `WekmbiUmSMa-LfttLegoZA`: Service succeeded, Infrastructure node Aborted, run Aborted. The error from the parent matrix node reaches the Infrastructure node and the environment rule applies, not the service rule.

08 Status summary

Behaviour	Cases	Status
Service & environment retry records N attempts on the right node	1, 4	PASS
Bare-string actions accepted and applied at both scopes	2, 3, 5	PASS
Stage rollback runs rollback steps on environment failure	6	PASS
Error matching: wildcard catches, narrow list does not	7, 8	PASS
Plan-time validation with NG wording; no retry in retry	9–12	PASS
Multi-deployment isolation, skip-queued, runtime inputs	13–16	PASS
K8s Initialize fails fast on errors other than "service not found"	16, 25	PASS
Priority and fallback: service rule first, stage rule when it does not match	17, 18	PASS
Pipeline rollback rolls back earlier stages	19	PASS
Inherited stage action is applied on Service and Infrastructure	20	PASS
Happy path unaffected	21	PASS
Missing infrastructure handled by the environment rule, as in NG	22–24	PASS

09 Not covered

Item	Why
Runtime test of every error type (authentication, connectivity, timeout, and so on)	These cannot be produced on demand at service or environment scope. Only `unknown` was run. All types map through the shared `NGFailureTypeV1` code.
Retry attempts in the execution view	UI gap, tracked as CDS-135093.
Schema-level rejection of disallowed actions at save time	Not changed in this work; rejection happens when the run starts.
Environment group with `on-failure`	Read from the code only; no pipeline with an environment group was run.

Pipeline cases ran on a local Harness stack against a local Kubernetes cluster, using the V1 pipeline API. Fixtures are in `test-test/CDS-121526-test/verify`, `demos` and `checks`.